

Callum Huntington

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Education

PhD in Mathematics and its Applications University of Naples Federico II, Italy
2023–present (expected conclusion in 2026)

- Thesis: *On joins, products, and structural operations of signed graphs and gain graphs*
- Supervisor: Francesco Belardo

MSc in Mathematics (with Distinction) University of Sheffield, United Kingdom
2020–2021

- Thesis: *Hyperbolic Geometry and Sandpiles*
- Supervisor: Paul Johnson

BSc in Mathematics (with Class One Honours) University of Sheffield, United Kingdom
2017–2020

Research Experience

PhD Researcher University of Naples Federico II, Italy
2023–present

- Research focus: spectral theory of signed graphs and gain graphs, with emphasis on structural operations (joins, products, corona constructions) and switching equivalence. Developed spectral methods using linear algebra and group representations for gain graphs over arbitrary groups.
- My research bridges spectral graph theory and algebraic combinatorics. My work has appeared in or is under review at the leading specialty journals for matrix theory (Linear Algebra and its Applications, Electronic Journal of Linear Algebra) as well as broad combinatorics outlets (Discrete Mathematics, Electronic Journal of Combinatorics).
- Collaborations with researchers at Niccolò Cusano University, Italy (S. Spessato, A. Donno, M. Cavaleri), University of Belgrade, Serbia (Z. Stanić), University of the Western Cape, South Africa (O. Waka), and Eindhoven University of Technology, The Netherlands (A. Abiad, A. Khramova).

Publications

Published

- C. Huntington. Spectra and balance of signed graphs obtained from vertex splittings. *Discussiones Mathematicae Graph Theory*, in press, 2026. DOI: 10.7151/dmgt.2643.
- C. Huntington, S. Spessato, F. Belardo. On H -joins of complex unit gain graphs and their stability. *Discrete Mathematics*, vol. 349, no. 7, pp. 115069, 2026. DOI: 10.1016/j.disc.2026.115069.
- C. Huntington. The $\dot{H}$ -join operation of signed graphs constrained by indexing maps. *RAIRO-Operations Research*, vol. 59, no. 5, pp. 2501–2515, 2025. DOI: 10.1051/ro/2025100.

- F. Belardo, C. Huntington. Signed graphs with strong (anti-)reciprocal eigenvalue property. *Special Matrices*, vol. 12, no. 1, pp. 20240017, 2024. DOI: 10.1515/spma-2024-0017.

Under review and in preparation

- C. Huntington, Z. Stanić, F. Belardo. Further spectral relations between oriented graphs and corresponding signed graphs. *Under review*.
- C. Huntington, O. Waka. Spectra of gain Q and R graphs via G -phases and group representations. *Under review*.
- C. Huntington, F. Belardo, S. Barik. Graphs with integer reciprocal eigenvalue properties. *Under review*.
- F. Belardo, A. Donno, C. Huntington, S. Spessato. A switching-stable corona product of signed graphs. *Under review*.
- C. Huntington. Spectral properties and balance of products derived from gain subdivision graphs via group representations. *In preparation*.
- A. Abiad, C. Huntington, A.P. Khramova. A switching method for signed and gain sun graphs. *In preparation*.
- C. Huntington. Quotient graphs and spectra of Leonardo grids. *In preparation*.

Conferences

- **Real-world Applications of Geometry and Algebra 2026**. Eindhoven, The Netherlands, June 2026.
- **Combinatorics 2026**. Naples, Italy, May 2026.
- **Mini Workshop of Topology and Algebra 2025**. Cape Town, South Africa, December 2025. Invited talk: *The H -join of complex unit gain graphs and their stability*.
- **Workshop on Spectral Graph Theory 2025**. Gramado, Brazil, October 2025. Contributed talk: *Spectral relations between oriented graphs and corresponding signed graphs*.
- **12th PhD Summer School in Discrete Mathematics**. Koper, Slovenia, September 2025. Contributed talk: *The H -join of complex unit gain graphs and their stability*.
- **CIME Summer School: Modern Perspectives in Approximation Theory**. Cetraro, Italy, July 2025.
- **IRGTADS Seminars**. Online, June 2025. Invited talk: *The H -join operation on signed graphs*.
- **Group Theory in Salerno**. Salerno, Italy, March 2025.
- **1st Chinese-SouthEastEuropean Conference on Discrete Mathematics and Applications**. Belgrade, Serbia, June 2024. Contributed talk: *Constructions of nonregular cospectral signed graphs*.
- **University of Pisa Summer School: Introduction to Machine Learning in Geosciences**. Pisa, Italy, July 2022.

Awards and Funding

- **Travel Grant for RAGA 2026** – Eindhoven University of Technology, €715, 2026.
- **Doctoral Programme Scholarship** – University of Naples Federico II, €16,243, 2023–2026.

Additional Experience

Warehouse Operative
2016–2023, 2026–present

Arthrex UK, Sheffield, United Kingdom

Other Skills and Qualifications

- **Languages:** English (native), Italian (B1).
- **Programming:** Basic knowledge of Python, SageMath, R, and MATLAB.
- **Refereeing activity:** Served as a peer reviewer for several international journals in graph theory and discrete mathematics.